

MANIKANDAN P

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Chennai / Bengaluru, India | Graduating June 2026 | Open to Relocation

PROFESSIONAL SUMMARY

Final-year Aerospace Engineering student at Hindustan Institute of Technology and Science (graduating June 2026) with multi-domain industry experience spanning aerostructure manufacturing (Tata Advanced Systems – Airbus C295), military helicopter systems & airworthiness (DRDO RCMA), and structural FEA (L&T Shipbuilding). Co-inventor of an Indian Patent in additive manufacturing for aerospace structures. Demonstrated technical depth across composites, FEA, space radiation analysis (43.9% TID reduction, CubeSat – ANSYS + NASA OLTARIS), and systems-level rover design (NASA HERC 2025 – Team Leader, 11-member team). Seeking a role at the intersection of structural engineering, advanced manufacturing, and aerospace systems.

EDUCATION

B.Tech – Aerospace Engineering Aug 2022 – Jun 2026
Hindustan Institute of Technology and Science, Chennai **CGPA: 7.68 / 10**
Core Domains: Aerospace Structures · Propulsion Systems · Composite Materials · Aeroelasticity

TECHNICAL SKILLS

CAD & Design	CATIA V5 (Part Design, Assembly, GSD, Drafting), SolidWorks, AutoCAD
Structural Analysis	ANSYS Mechanical (Static Structural, Thermal, Linear Buckling, Modal), Classical Lamination Theory (CLT)
Space Systems	NASA OLTARIS (TID / Radiation Transport), CubeSat Structural Design, Orbital Environment Modelling
Composites & Mfg	Composite Hand Layup, Co-Curing, Aerostructure Assembly, FDM / SLS Additive Manufacturing, GD&T, SAP ERP
Programming	Python (NumPy, SciPy, Data Processing), MATLAB, $\text{\LaTeX}$
Standards	AS9100, CEMILAC Airworthiness, RCMA Military Certification, Aerospace Quality Inspection

INTERNSHIP EXPERIENCE

Intern Trainee – Airbus C295 Structural Manufacturing Jan 2026 – Mar 2026
Tata Advanced Systems Limited (TASL), Bengaluru (Full-time, On-site)

- Executing primary aerostructure manufacturing for the **Airbus C295 military transport aircraft** – one of India’s highest-volume defence aerospace production programs (HAL-Airbus JV).
- Applying composite layup, co-curing cycles, and film/paste adhesive bonding to structural panel assemblies; maintaining fibre orientation, ply drop, and bond-line compliance to engineering drawings.
- Managing component traceability and work order flow via **SAP ERP**; supporting AS9100 audit documentation and non-conformance reporting (NCR) processes.
- Performing in-process dimensional inspection and tolerance verification; documenting corrective actions in alignment with aerospace quality management systems.

Engineering Intern – Structural FEA & Rotating Systems Jun 2025 – Jul 2025
L&T Shipbuilding Limited, Kattupalli (On-site)

- Performed static structural **FEA on mechanical joint assemblies** using CATIA V5; identified stress concentration regions and assessed failure modes under marine loading.
- Conducted hand calculations for structural integrity and fatigue assessment of rotating components; cross-validated results with FEA simulation output.
- Documented manufacturing process standards and quality inspection procedures for precision-engineered rotating systems.

Intern – Helicopter Systems, Airworthiness & Structural Integration Dec 2024 – Jan 2025
DRDO, RCMA Helicopter Division (under CEMILAC), Bengaluru (On-site)

- Studied structural and systems integration of **ALH (Dhruv)**, **LCH**, and **LUH** helicopter platforms: Main Gear Box (MGB), Anti-Resonant Isolating System (ARIS), tail rotor drive shaft, and transmission assemblies.
- Reviewed CEMILAC and RCMA military airworthiness certification framework; traced certification lifecycle from design through flight clearance.

- Observed structural inspection (NDT), CNC machining, and **3DNX-based design workflows** at Systems Integration Lab (SIL).

KEY TECHNICAL PROJECTS

CubeSat Payload Structural Optimisation – LEO Radiation Shielding 2025–2026
ANSYS Mechanical (FEA) · NASA OLTARIS · Python · MATLAB

- Designed a hybrid **CFRP–HDPE–Aluminium** shielding enclosure for a CubeSat payload in 600 km SSO; achieved **43.9% reduction in Total Ionising Dose (TID)** versus a 2 mm monolithic aluminium baseline – validated via NASA OLTARIS radiation transport analysis.
- Performed static structural, thermal, and combined loading analyses in **ANSYS FEM**; conducted parametric trade studies across 6 multi-material shielding architectures balancing mass, TID, and stiffness.
- Modelled orbital proton/electron flux environment and South Atlantic Anomaly (SAA) contribution to total dose; generated dose-depth curves for candidate material stacks.

NASA HERC 2025 – Planetary Rover Chassis Design & Systems Integration 2024–2025
CATIA V5 · Structural Hand Calculations · Drivetrain Analysis

- Led **11-member interdisciplinary team** as Student Team Leader for NASA Human Exploration Rover Challenge 2025 (Team Novanix) – international university-level planetary rover competition.
- Designed foldable truss-frame rover chassis (unfolded: $2600 \times 1600 \times 1250$ mm) in **Aluminium Alloy 6061** using CATIA V5; integrated H-joint folding mechanism with pin-lock retention for transport compliance.
- Calculated drivetrain performance: gear ratio, torque demand, and traverse speed for the 1,126 m course – achieved **2.34 m/s** design traverse speed.
- Oversaw full system integration: A-arm suspension, Ackerman joystick steering, chain-drive drivetrain, and 4-wheel mechanical disc braking; coordinated PDR/CDR deliverables.

Composite Aircraft Radome – Fibre Draping and CLT Displacement Analysis 2025
MATLAB · Classical Lamination Theory (CLT)

- Implemented CLT in MATLAB to compute laminate [ABD] stiffness matrix and predict structural displacement; demonstrated **39% displacement reduction** using astroquartz over E-glass fibre.
- Quantified **25% lower shear angle deformation** in woven vs. UD fibre configurations, reducing inter-ply debonding risk over doubly-curved radome geometry.
- *Manuscript submitted – International Conference on Smart Materials & Technology (ICSMT 2025).*

PATENT

A Process for Preparing Sandwich Core using Additive Manufacturing
Indian Patent Application No. 202541064323 A Co-inventor Filed: July 2025 · Indian Patent Office
Proposes FDM and SLS-based lattice sandwich core geometries for improved stiffness-to-weight ratio in lightweight aerospace structural panels – applicable to aerostructure and space component manufacturing.

CERTIFICATIONS

- SPARC – Analysis and Statistical Modelling of Space Data, MCNS MAHE (2025)
- SPARC III – Fundamentals of Space Radiation, MCNS MAHE (2024)
- Advanced Diploma in Python Programming (2021)

LEADERSHIP & EXTRACURRICULAR

Design & Media Head | Web Administrator | Junior Researcher Sep 2022 – Present
Nebula Astro Club, Hindustan Institute of Technology and Science
Progressed from Junior Researcher to Design & Media Head over 3+ years; managed club web presence as Web Administrator for 1.5 years.

LANGUAGES

English – Professional Proficiency | Tamil – Native Proficiency